PRAKTIKUM SISTEM OPERASI

MODUL IV

Membaca Source Code Xinu

Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

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JURNAL

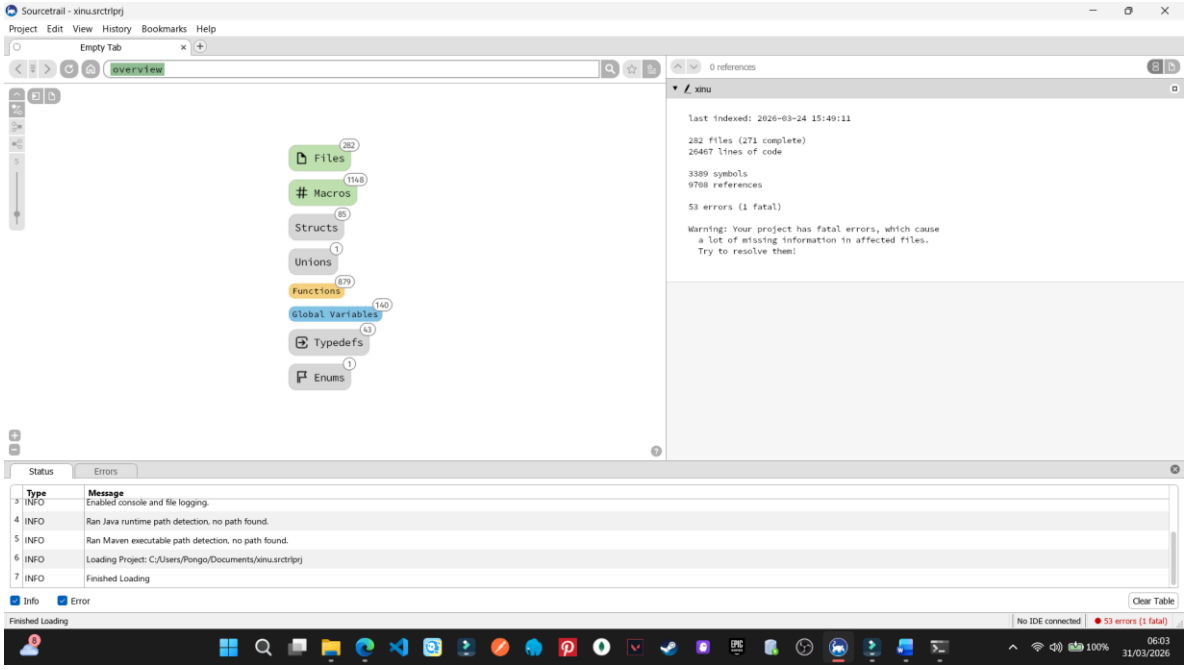
1. Apa nama image yang dihasilkan setelah melakukan kompilasi pada Xinu? Berapa ukuran file tersebut? Ada pada folder apa file image tersebut?

   Jawaban: Xinu.elf. Dibuat setelah proses make clean -> make lalu di ls.

```
xinu@xinu-develop-end:~$ ls -lh ~/xinu/compile/xinu.elf
-rwxr-xr-x 1 xinu xinu 156K Mar 26 07:37 /home/xinu/xinu/compile/xinu.elf
```

2. Membaca source code Xinu

   Jawaban: Import file tar.gz beserta include path-nya.



3. Carilah struktur data dari proses pada Xinu OS. Struktur data proses ada pada file apa? Informasi apa saja yang disimpan dalam struktur data tersebut?

   Jawaban: Command process.h didalam xinu/include.

```
xinu@xinu-develop-end:~$ grep -r "struct procent" ~/xinu/include/
/home/xinu/xinu/include/process.h:struct procent {              /* Entry
 table          */
```

```
xinu@xinu-develop-end:~$ cat ~/xinu/include/process.h
/* process.h - isbadpid */

/* Maximum number of processes in the system */

#ifndef NPROC
#define NPROC           8
#endif

/* Process state constants */

#define PR_FREE         0       /* Process table entry is unused        *
#define PR_CURR         1       /* Process is currently running         *
#define PR_READY        2       /* Process is on ready queue            *
#define PR_RECV         3       /* Process waiting for message          *
#define PR_SLEEP        4       /* Process is sleeping                  *
#define PR_SUSP         5       /* Process is suspended                 *
#define PR_WAIT         6       /* Process is on semaphore queue        *
#define PR_RECTIM       7       /* Process is receiving with timeout    *

/* Miscellaneous process definitions */

#define PNMLEN          16      /* Length of process "name"             *
#define NULLPROC        0       /* ID of the null process               *

/* Process initialization constants */
```

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#define PNMLEN          16      /* Length of process "name"
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/* Process initialization constants */

#define INITSTK         65536   /* Initial process stack size
#define INITPRIO        20      /* Initial process priority
#define INITRET         userret /* Address to which process returns

/* Inline code to check process ID (assumes interrupts are disabled)

#define isbadpid(x)     ( ((pid32)(x) < 0) || \
                          ((pid32)(x) >= NPROC) || \
                          (proctab[(x)].prstate == PR_FREE))

/* Number of device descriptors a process can have open */

#define NDESC           5       /* Must be odd to make procent 4N bytes

/* Definition of the process table (multiple of 32 bits) */

struct procent {                /* Entry in the process table
        uint16  prstate;        /* Process state: PR_CURR, etc.
        pri16   prprio;         /* Process priority
```

```
struct procent {                /* Entry in the process table
        uint16  prstate;        /* Process state: PR_CURR, etc.
        pri16   prprio;         /* Process priority
        char    *prstkptr;      /* Saved stack pointer
        char    *prstkbase;     /* Base of run time stack
        uint32  prstklen;       /* Stack length in bytes
        char    prname[PNMLEN]; /* Process name
        sid32   prsem;          /* Semaphore on which process waits
        pid32   prparent;       /* ID of the creating process
        umsg32  prmsg;          /* Message sent to this process
        bool8   prhasmsg;       /* Nonzero iff msg is valid
        int16   prdesc[NDESC];  /* Device descriptors for process
};

/* Marker for the top of a process stack (used to help detect overflow)
#define STACKMAGIC      0x0A0AAAA9

extern  struct  procent proctab[];
extern  int32   prcount;        /* Currently active processes
extern  pid32   currpid;        /* Currently executing process
xinu@xinu-develop-end:~$
```

4. Mengubah welcome banner pada Xinu
   a. Carilah file yang menyimpan banner Xinu!

      Jawaban: grep -r "Welcome to Xinu" ~/xinu/include/

```
xinu@xinu-develop-end:~$ grep -r "Welcome to Xinu" ~/xinu/include/
/home/xinu/xinu/include/shell.h:#define SHELL_STRTMSG   "Welcome to Xinu!"
message          */
```

   b. Carilah file yang menampilkan banner Xinu!

      Jawaban: grep -r "SHELL_STRTMSG" ~/xinu/shell/

```
xinu@xinu-develop-end:~$ grep -r "SHELL_STRTMSG" ~/xinu/shell/
/home/xinu/xinu/shell/shell.c:  fprintf(dev, "%s\n\n", SHELL_STRTMSG);
```

   c. Ubahlah welcome banner Xinu

```
/* Shell banner (assumes VT100) */

#define SHELL_BAN0      "\033[31;1m"
#define SHELL_BAN1      "----------------Bangjoule-----------------"
#define SHELL_BAN2      "   __    __   _____    _   _    _    _    "
#define SHELL_BAN3      "   \\ \\  / /  |__ __|  | \\ | |  | |  | |  "
#define SHELL_BAN4      "    \\ \\/ /     | |    |  \\| |  | |  | |  "
#define SHELL_BAN5      "    / /\\ \\     _| |_   | \\   |  | |  | |  "
#define SHELL_BAN6      "   / /  \\ \\   |     |  | | \\ |  \\  --  /"
#define SHELL_BAN7      "   --    --   -----    -   -     ----     "
#define SHELL_BAN8      "---------------2311104010-----------------"
#define SHELL_BAN9      "\033[0m\n"

/* Messages shell displays for user */

#define SHELL_PROMPT    "xsh $ "        /* Command prompt
#define SHELL_STRTMSG   "\n--- Nama ---\nBangjoule\n--- NIM ---\n231110
#define SHELL_EXITMSG   "Shell closed\n"/* Shell exit message
#define SHELL_SYNERRMSG "Syntax error\n"/* Syntax error message
#define SHELL_CREATEMSG "Cannot create process\n"/* command error
#define SHELL_INERRMSG  "Cannot open file %s for input\n" /* Input err
#define SHELL_OUTERRMSG "Cannot open file %s for output\n"/* Output err
                                [ Read 77 lines ]
^G Get Help   ^O WriteOut   ^R Read File  ^Y Prev Page  ^K Cut Text
^X Exit       ^J Justify    ^W Where Is   ^V Next Page  ^U UnCut Text
```

   d. Setelah melakukan perubahan, masuk ke minicom untuk melihat hasil perubahan terakhir.

```
Obtained IP address  10.0.3.15   (0x0a00030f)



----------------Bangjoule-----------------
   __    __   _____   _   _    _    _
   \ \  / /  |__ __|  | \ | |  | |  | |
    \ \/ /     | |    |  \| |  | |  | |
    / /\ \    _| |_   | \    |  | |  | |
   / /  \ \  |     |  | | \ |  \  --  /
   --    --   -----    -   -     ----
---------------2311104010-----------------



--- Nama ---
Bangjoule
--- NIM ---
2311104010
Welcome to Xinu!
```